CS1910 Computer Science I • Problem Solving and Debugging **Partner workbook**

Ants on a Stick

Start with a partner in lecture, finish at home • about 60 minutes

Name: _____ **Partner:** _____

This PDF can be filled in: click in an answer space to type, or print it and write. Draw diagrams with your PDF viewer's pen or markup tool, or on paper.

What this workbook practises

A puzzle that looks like it needs a complicated simulation turns out to have a short, elegant answer, *if* you follow the problem-solving process: **Understand** → **Design** → **Implement** → **Evaluate**. You will pin down exactly what the question asks, work small cases, spot the pattern that makes collisions irrelevant, and then check your answer against cases you worked out by hand. The Colab notebook `CS1910_Problem-Solving-Debugging-Ants_Handout.ipynb` has a simulator that animates every case in this workbook.

1 Understand

about 15 minutes

The problem, as first given

Ants march on a stick. They travel in one direction, but if they bump into another ant both ants reverse direction and walk the other way. If they reach the end of the stick they fall off. Ants travel at 1 cm/s. What is the minimum or maximum amount of time until the stick has no ants on it?
Input: the number of ants and the starting location of each ant. *Output:* the minimum and maximum possible time for the stick to empty.

Before designing anything, find out what you **don't** know.

1. What are the inputs, and what is the output?

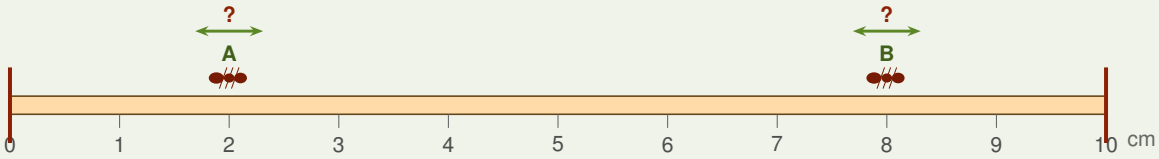
2. The output is the minimum and maximum **possible** time. Possible over *what*? What information about each ant is deliberately missing?

3. List at least **four** questions you would ask before you start designing. (Think about the stick, the positions, collisions, and unusual inputs.)

4. Which details of the story can be ignored completely? (Abstraction: what does not affect the answer?)

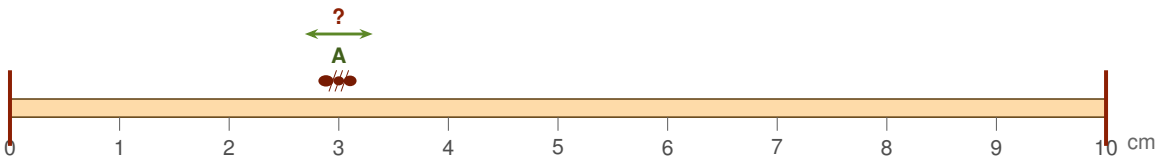
The clarified problem

- The stick is **10 cm** long. Positions are measured in cm from the left end, so the ends are at **0** and **10**.
- Each ant starts at a different **whole-number** position from 1 to 9, facing left or right, but **we do not know which**.
- Ants walk at **1 cm/s**. When two ants meet, both turn around instantly. An ant falls off the moment it reaches an end.
- **Input:** a string with one digit per ant, e.g. `"28"` means ants at 2 cm and 8 cm.
- **Output:** the **earliest** and **latest** times (in seconds) the stick could become empty, over every possible choice of directions. With no ants, both are 0.



The input `"28"`: two ants, each facing an unknown direction.

Start with the simplest case: one ant



One ant at 3 cm. There is nobody to collide with.

If it walks **left**, it falls off after _____ seconds

If it walks **right**, it falls off after _____ seconds

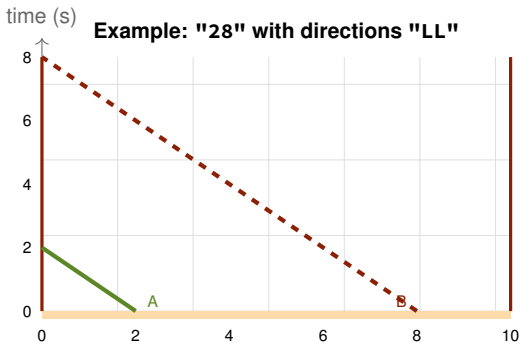
So the **earliest** time the stick can be empty is _____ seconds

and the **latest** time is _____ seconds.

2 Explore small cases

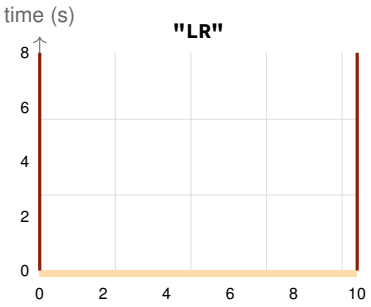
about 10 minutes

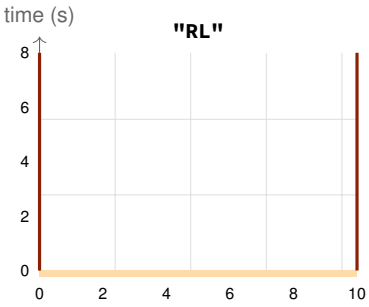
Two ants are harder to do in your head, because they might collide. A **space-time diagram** helps: position goes across, time goes *up*, and each ant's path is a line. An ant moving at 1 cm/s makes a diagonal line; a collision is where two lines meet, and an ant falls off when its line reaches a red end.

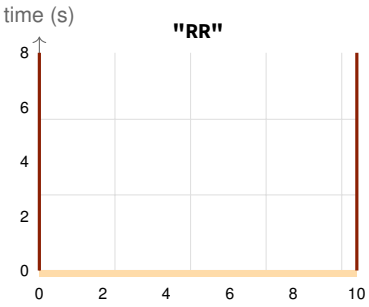


In the example both ants walk left: A falls off after 2 s and B after 8 s, so the stick is empty after **8 s**.

5. Draw the space-time diagrams for the other three choices of direction for `"28"`. Mark any collision with a dot.







6. Fill in the table.

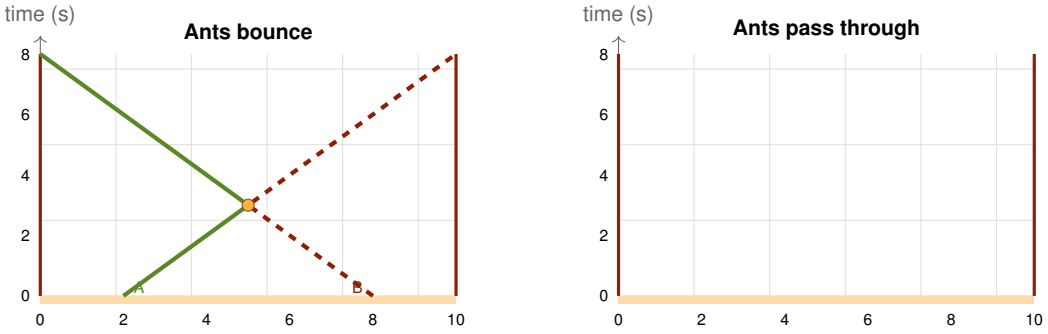
Directions	Collision?	Stick empty after (s)
`"LL"`	no	8
`"LR"`		
`"RL"`		
`"RR"`		

7. What are the earliest and latest times for `"28"`? Which directions give each?

3 Recognize the pattern

about 10 minutes

On the left are the paths for `"28"` with directions `"RL"`, where the ants **bounce**. On the right, draw the paths as if the ants were ghosts that **pass straight through** each other.



8. Ignore which ant is which, and look only at the *shape* of the lines. What do you notice about the two pictures?

9. Does it matter *when the stick becomes empty* whether ants bounce or pass through? Why?

10. If ants pass through each other, how long does an ant at position p take to fall off walking left? Walking right?

11. The stick is empty when the **last** ant falls off. To make that happen as **early** as possible, which way should each ant walk? Write a rule for the earliest time.

12. To make it happen as **late** as possible, what is the best choice? Write a rule for the latest time.

13. Your friend says "the earliest time is the *smallest* nearer-end distance of any ant." Test this on `"15"`. Are they right?

Key ideas: the ants

- Bouncing ants and passing-through ants make the **same picture**, so collisions never change when the stick is empty.
- Each ant then takes p seconds (left) or $10 - p$ seconds (right), on its own.
- **Earliest** time $=$ the largest of $\min(p,\ 10 - p)$ over all ants. **Latest** time $=$ the largest of $\max(p,\ 10 - p)$ over all ants.

4 Design

about 5 minutes

You will write two functions, `earliest_empty(positions)` and `latest_empty(positions)`. Each takes the input string and returns a whole number of seconds.

14. Write pseudocode for `earliest_empty`. Then check it by hand against `"28"`, `"15"` and `""`. How does `latest_empty` differ?

5 Implement

about 10 minutes

15. Write both functions in Python. You may use `min(a, b)` and `max(a, b)`, or `if` statements.

6 Evaluate

about 10 minutes

A fast answer is only useful if it is **right**. Test your functions on cases whose answers you can work out a different way: by drawing a space-time diagram, or with the brute-force simulator in the notebook (`simulate_all`), which tries every possible choice of directions.

16. Complete the test table. Work out the expected answers *before* you run your code, then run it and tick the last column.

Input	Earliest	Latest	Why this test?	OK?
`"5"`				
`"9"`				
`"15"`				
`"123456789"`				
`""`				

17. The brute-force simulator tries every combination of directions. How many simulations is that for 2 ants? 9 ants? About how many for 100 ants on a 1 m stick? How many steps does your function need for 100 ants?

18. Explain, in one or two sentences, why we do not need to simulate collisions at all.

Extension: any length of stick

Add a parameter `length=10` to both functions and use it instead of `10`. What are the earliest and latest times for `"28"` on a 12 cm stick? _____

✓ Before you leave

about 5 minutes

Tick each statement you can honestly agree with.

- ☐ I can find what is ambiguous in a problem statement and ask questions to pin it down.
- ☐ I can read and draw a space-time diagram.
- ☐ I can explain why collisions do not change when the stick becomes empty.
- ☐ I can explain why the earliest time is the *largest* nearer-end distance, not the smallest.
- ☐ I can test a fast solution against answers worked out another way.

19. Which computational-thinking skill helped most here: abstraction, decomposition, pattern recognition, algorithm design, or validation? Why?

20. At what point did your understanding of the problem change? What changed it?